

Topic 1: Basic Linux Commands

1. **Pwd** ~ Display the current working directory

```
saisharan134@APL-5CD1366K5L:~$ pwd  
/home/saisharan134
```

2. **Clear** ~ clear the terminal screen

```
saisharan134@APL-5CD1366K5L:~$
```

3. **History** ~ displays previously executed commands

```
133  pwd  
134  clear  
135  history  
saisharan134@APL-5CD1366K5L:~$
```

4. **Whoami** ~ display the current logged-in username

```
saisharan134@APL-5CD1366K5L:~$ whoami  
saisharan134
```

5. **Date** ~ display the current system date and time

```
saisharan134@APL-5CD1366K5L:~$ date  
Tue Aug 11 04:05:53 UTC 2026
```

6. **Hostname** ~ Display the system hostname

```
saisharan134@APL-5CD1366K5L:~$ hostname  
APL-5CD1366K5L
```

7. **Uname -a** ~ Display operating system operation

```
saisharan134@APL-5CD1366K5L:~$ uname -a  
Linux APL-5CD1366K5L 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DY  
NAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux
```

8. **Exit** ~ exit the terminal

```
saisharan134@APL-5CD1366K5L:~$ exit|
```

TOPIC 2 ~ DIRECTORY MANAGEMENT

1. **Ls** ~ list files and directories

```
saisharan134@APL-5CD1366K5L:~$ ls  
EighthProgramme  FourthProgramme.c  SeventhProgramme  
EighthProgramme.c  FristProgramme.c   SeventhProgramme.c  
Example           Hello              SixthProgramme
```

2. Ls -a ~ display detailed file information

```
saisharan134@APL-5CD1366K5L:~$ ls -l
total 232
-rwxr-xr-x 1 saisharan134 saisharan134 16008 Aug  4 14:11 EighthProgramme
-rw-r--r-- 1 saisharan134 saisharan134   160 Aug  4 14:11 EighthProgramme.c
drwxr-xr-x 2 saisharan134 saisharan134  4096 Jul 28 04:06 Example
-rw----- 1 saisharan134 saisharan134   864 Jul 31 09:14 Example.c.save
-rwxr-xr-x 1 saisharan134 saisharan134 16000 Aug  4 14:05 FifthProgramme
```

```
saisharan134@APL-5CD1366K5L:~$ ls -a
.      .motd_shown      FirstProgramme      Hello.c.save.1      SixthProgramme.c
..     .profile         FirstProgramme.c    NinthProgramme      ThirdProgramme
.bash_history EighthProgramme     FirstProgramme.c    NinthProgramme.c    ThirdProgramme.c
.bash_logout EighthProgramme.c   FourthProgramme     Programme           hello
.bashrc     Example           FourthProgramme.c   SecondProgramme     hello.c
.cache      Example.c.save     FristProgramme.c    SecondProgramme.c   s2SamplePgrms
```

3. Cd ~ change the current directory

```
saisharan134@APL-5CD1366K5L:~$ cd
saisharan134@APL-5CD1366K5L:~$
```

4. Cd .. ~ move to parent directory

```
saisharan134@APL-5CD1366K5L:~$ cd ..
saisharan134@APL-5CD1366K5L:/home$
```

5. mkdir ~ create a new directory

```
saisharan134@APL-5CD1366K5L:~$ mkdir OSSPLAB
```

6. rmdir ~ remove a **directory**

```
saisharan134@APL-5CD1366K5L:~$ rmdir OSSPLAB
```

7. find FileName.c ~ search for files and directories

```
saisharan134@APL-5CD1366K5L:/home$ find Hello.c
```

8. locate ~ Quickly locate files

```
saisharan134@APL-5CD1366K5L:/home$ locate Hello.c
```

3.FILE MANAGEMENT

1.touch ~ Create new empty file

```
saisharan134@APL-5CD1366K5L:~$ touch KL.c
```

2. cat ~ display file contents

```
saisharan134@APL-5CD1366K5L:~$ cat Hello.c
#include<stdio.h>

int main()
{
    printf("Hello Linux\n");
    return 0;
}
```

3. cp ~ copy files or directories

```
saisharan134@APL-5CD1366K5L:~$ cp Hello.c FirstProgramme.c
```

4. mv ~ move or rename files

```
saisharan134@APL-5CD1366K5L:~$ mv Hello.c hello.c
saisharan134@APL-5CD1366K5L:~$
```

5. rm ~ remove file

```
saisharan134@APL-5CD1366K5L:~$ rm hello.c
```

6. head ~ display first lines of a file

```
saisharan134@APL-5CD1366K5L:~$ head FirstProgramme.c
#include<stdio.h>

int main()
{
    printf("Hello Linux\n");
    return 0;
}
```

7. tail ~ display last lines of the file

```
saisharan134@APL-5CD1366K5L:~$ tail FirstProgramme.c
#include<stdio.h>

int main()
{
    printf("Hello Linux\n");
    return 0;
}
```

8.more ~ view a file page by page

```
saisharan134@APL-5CD1366K5L:~$ more FirstProgramme.c
#include<stdio.h>

int main()
{
    printf("Hello Linux\n");
    return 0;
}
```

9.less ~ Scroll through a file

```
#include<stdio.h>

int main()
{
    printf("Hello Linux\n");
    return 0;
}
FirstProgramme.c (END)
```

10.wc ~ count lines, words and characters

```
saisharan134@APL-5CD1366K5L:~$ wc FirstProgramme.c
 7  9 83 FirstProgramme.c
```

11.diff ~ compare two files

```
saisharan134@APL-5CD1366K5L:~$ diff FirstProgramme.c SecondProgramme.c
1c1
< #include<stdio.h>
---
> #include <stdio.h>
5,6c5,12
<     printf("Hello Linux\n");
<     return 0;
---
>     char name[30];
>
>     printf("Enter your name: ");
>     scanf("%s", name);
>
>     printf("Welcome %s\n", name);
>
>     return 0;
saisharan134@APL-5CD1366K5L:~$ |
```

12.sort ~ sort file contents

```
saisharan134@APL-5CD1366K5L:~$ sort FirstProgramme.c

#include<stdio.h>
int main()
{
}

printf("Hello Linux\n");
return 0;
```

13. uniq ~ remove duplicate files

```
saisharan134@APL-5CD1366K5L:~$ uniq FirstProgramme.c
#include<stdio.h>

int main()
{
    printf("Hello Linux\n");
    return 0;
}
```

14. grep ~ search for text inside a file

```
saisharan134@APL-5CD1366K5L:~$ grep main FirstProgramme.c
int main()
```

4. TOPIC ~ FILE EDITING

1. nano ~ create or edit a file

```
saisharan134@APL-5CD1366K5L:~$ nano Hello.c
saisharan134@APL-5CD1366K5L:~$ |
```

2. vi ~ edit a file using vim editor

```
#include<stdio.h>

int main()
{
    printf("Hello Linux\n");
    return 0;
}
```

5. C PROGRAM COMPILATION AND EXECUTION

1. gcc ~ compile a C program

```
saisharan134@APL-5CD1366K5L:~$ gcc FirstFile.c -o file
saisharan134@APL-5CD1366K5L:~$ |
```

2. ./program ~ Execute the compiled program

```
saisharan134@APL-5CD1366K5L:~$ ./file
Hello Linux
```

3. which ~ Locate the executable of a command

```
saisharan134@APL-5CD1366K5L:~$ which gcc
/usr/bin/gcc
```

4. whereis ~ Locate binary, source and manual pages

```
saisharan134@APL-5CD1366K5L:~$ whereis gcc
gcc: /usr/bin/gcc /usr/lib/gcc /usr/libexec/gcc /usr/share/gcc /usr/share/man/man1/gcc.1.gz
```

5. man ~ display manual pages

```
GCC(1)                                GNU                                GCC(1)

NAME
    gcc - GNU project C and C++ compiler

SYNOPSIS
    gcc [-c|-S|-E] [-std=standard]
        [-g] [-pg] [-Olevel]
        [-Wwarn...] [-Wpedantic]
        [-Idir...] [-Ldir...]
        [-Dmacro[=defn]...] [-Umacro]
        [-foption...] [-mmachine-option...]
        [-o outfile] [@file] infile...

    Only the most useful options are listed here; see below for the
    remainder.  g++ accepts mostly the same options as gcc.

DESCRIPTION
    When you invoke GCC, it normally does preprocessing,
    compilation, assembly and linking.  The "overall options" allow
    you to stop this process at an intermediate stage.  For example,
    the -c option says not to run the linker.  Then the output
    consists of object files output by the assembler.

    Other options are passed on to one or more stages of processing.
    Some options control the preprocessor and others the compiler.
```

6.PROCESS MANAGEMENT

1. ps ~ display running processes

```
saisharan134@APL-5CD1366K5L:~$ ps
  PID TTY          TIME CMD
 1913 pts/0        00:00:00 bash
 2281 pts/0        00:00:00 ps
```

2. ps -ef ~ display all running processes

```
saisharan134@APL-5CD1366K5L:~$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1         0  0  04:22 ?        00:00:02 /sbin/init
root           2         1  0  04:22 hvc0    00:00:00 /init
root           7         2  0  04:22 hvc0    00:00:00 plan9 --control-sock
root          53         1  0  04:22 ?        00:00:00 /usr/lib/systemd/sys
systemd+      88         1  0  04:22 ?        00:00:00 /usr/lib/systemd/sys
root          96         1  0  04:22 ?        00:00:04 /usr/lib/systemd/sys
root         120         1  0  04:22 ?        00:00:00 /bin/sh /usr/lib/sys
root         121         1  0  04:22 ?        00:00:00 /usr/sbin/cron -f -P
message+     122         1  0  04:22 ?        00:00:00 @dbus-daemon --syste
root         128         1  0  04:22 ?        00:00:00 /usr/bin/python3 /us
root         167         1  0  04:22 ?        00:00:00 /usr/lib/systemd/sys
syslog       200         1  0  04:22 ?        00:00:00 /usr/sbin/rsyslogd -
_chrony      237        120  0  04:22 ?        00:00:00 /usr/sbin/chronyd -n
_chrony      245        237  0  04:22 ?        00:00:00 /usr/sbin/chronyd -n
root         284         1  0  04:22 tty1    00:00:00 /usr/sbin/agetty --n
root         286         1  0  04:22 ?        00:00:00 /usr/bin/python3 /us
root         341         2  0  04:22 ?        00:00:00 login -- saisharan13
saishar+     392         1  0  04:22 ?        00:00:00 /usr/lib/systemd/sys
```

3. top ~ monitor running processes

	PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
	2313	saishar+	20	0	8160	6068	3976	R	0.3	0.1	0:00.02	top
	1	root	20	0	24236	15348	11512	S	0.0	0.2	0:02.25	systemd
	2	root	20	0	3180	2208	2072	S	0.0	0.0	0:00.04	init-syste+
	7	root	20	0	3180	2048	1940	S	0.0	0.0	0:00.00	init
	53	root	19	-1	50424	16888	15604	S	0.0	0.2	0:00.52	systemd-jo+
	88	systemd+	20	0	22540	14656	12036	S	0.0	0.2	0:00.34	systemd-re+
	96	root	20	0	35320	12364	9316	S	0.0	0.2	0:04.20	systemd-ud+
	120	root	20	0	2888	2012	1884	S	0.0	0.0	0:00.14	chronyd-st+
	121	root	20	0	4472	3188	2940	S	0.0	0.0	0:00.06	cron
	122	message+	20	0	8788	5492	4824	S	0.0	0.1	0:00.18	dbus-daemon
	128	root	20	0	44092	29684	14540	S	0.0	0.4	0:00.71	networkd-d+
	167	root	20	0	18436	9468	8224	S	0.0	0.1	0:00.28	systemd-lo+
	200	syslog	20	0	220548	5824	4504	S	0.0	0.1	0:00.23	rsyslogd
	237	_chrony	20	0	20420	6468	5692	S	0.0	0.1	0:00.21	chronyd
	245	_chrony	20	0	12092	2484	1844	S	0.0	0.0	0:00.00	chronyd
	284	root	20	0	5402	2780	2596	S	0.0	0.0	0:00.01	agetty

4. kill ~ terminate a process

```
saisharan134@APL-5CD1366K5L:~$ kill 1234
-bash: kill: (1234) - No such process
```

5. kill -9 ~ Forcefully terminate a process

```
saisharan134@APL-5CD1366K5L:~$ kill -9
kill: usage: kill [-s sigspec | -n signum | -sigspec] pid | jobspec ... or kill -l
[sigspec]
```

7. JOB CONTROL

1. jobs ~ Display background jobs

```
saisharan134@APL-5CD1366K5L:~$ jobs
saisharan134@APL-5CD1366K5L:~$
```

2. bg ~ Resume a job in the background

```
saisharan134@APL-5CD1366K5L:~$ bg %1
-bash: bg: %1: no such job
```

3. fg ~ Bring a job to the foreground

```
saisharan134@APL-5CD1366K5L:~$ fg %1
-bash: fg: %1: no such job
```

4. & - Run a process in the background

```
saisharan134@APL-5CD1366K5L:~$ ./file &
[1] 2404
Hello Linux
[1]+  Done                  ./file
```

8. FILE PERMISSIONS AND LINKS

1. chmod ~ Change file permissions

```
saisharan134@APL-5CD1366K5L:~$ chmod +x script.sh
```

2. chmod 755 ~ Set read, write and execute permissions

```
saisharan134@APL-5CD1366K5L:~$ chmod 755
```

3. ln ~ create a hard link

```
saisharan134@APL-5CD1366K5L:~$ ln FirstProgramme.c Hello.c
```

4. ln -s ~ create a symbolic link

```
saisharan134@APL-5CD1366K5L:~$ ln -s FirstProgramme shortcut
saisharan134@APL-5CD1366K5L:~$
```

9. DISK AND MEMORY INFORMATION

1. df -h ~ display disk space usage

```
saisharan134@APL-5CD1366K5L:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
none            3.9G   0 3.9G   0% /usr/lib/modules/6.18.33.2-microsoft-standard-WSL2
none            3.9G  4.0K 3.9G   1% /mnt/wsl
drivers          224G   92G 133G  41% /usr/lib/wsl/drivers
/dev/sdd        1007G   1.7G 955G   1% /
none            3.9G   64K 3.9G   1% /mnt/wslg
none            3.9G   0 3.9G   0% /usr/lib/wsl/lib
rootfs          3.8G  2.8M 3.8G   1% /init
none            3.9G  500K 3.9G   1% /run
none            3.9G   0 3.9G   0% /run/lock
none            3.9G   0 3.9G   0% /run/shm
none            3.9G   80K 3.9G   1% /mnt/wslg/versions.txt
none            3.9G   80K 3.9G   1% /mnt/wslg/doc
C:\              224G   92G 133G  41% /mnt/c
D:\              252G  2.8G 250G   2% /mnt/d
tmpfs            3.9G   0 3.9G   0% /tmp
none            1.0M   0 1.0M   0% /run/credentials/systemd-journald.service
none            1.0M   0 1.0M   0% /run/credentials/systemd-resolved.service
none            1.0M   0 1.0M   0% /run/credentials/getty@tty1.service
tmpfs            779M  12K 779M   1% /run/user/1000
```

2. du -sh ~ display directory size

```
saisharan134@APL-5CD1366K5L:~$ du -sh
396K .
```

3. free -h ~ display memory usage (Linux)

```
saisharan134@APL-5CD1366K5L:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:           7.6Gi         520Mi        6.9Gi         3.5Mi         310Mi        7.1Gi
Swap:          2.0Gi           0B         2.0Gi
```

10.USER AND ENVIRONMENT MANAGEMENT

1.id ~ display user and group IDs

```
saisharan134@APL-5CD1366K5L:~$ id
uid=1000(saisharan134) gid=1000(saisharan134) groups=1000(saisharan134),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),100(users)
```

2.env ~ display environment variables

```
saisharan134@APL-5CD1366K5L:~$ env
SHELL=/bin/bash
WSL2_GUI_APPS_ENABLED=1
WSL_DISTRO_NAME=Ubuntu
NAME=APL-5CD1366K5L
PWD=/home/saisharan134
LOGNAME=saisharan134
HOME=/home/saisharan134
LANG=C.UTF-8
WSL_INTEROP=/run/WSL/1909_interop
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33:cd=40;33:or=40;31:mi=00:su=37;41:sg=30;43:ca=00:tw=30;42:ow=34;4
2:st=37;44:ex=01;32:*.tar=01;31:*.tgz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lz4=01;31:*.lzh=01;31:*.lzma=01;31:*.tlz=01;31:*.txz=0
1;31:*.tzo=01;31:*.t7z=01;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lrz=01;31:*.lz=01;31:*.lzo=01;31:*.xz=01;31:*.zst=01;31:*.tzt=01;31:*.bz
2=01;31:*.bz=01;31:*.tbz=01;31:*.tbz2=01;31:*.tz=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.ear=01;31:*.sar=01;31:*.rar=01;31:*.alz=01;
31:*.ace=01;31:*.zoo=01;31:*.cpio=01;31:*.7z=01;31:*.rz=01;31:*.cab=01;31:*.wim=01;31:*.swm=01;31:*.dwm=01;31:*.esd=01;31:*.avif=01;35:*.jpg=01;35:*.
jpeg=01;35:*.mjpg=01;35:*.mjpeg=01;35:*.gif=01;35:*.bmp=01;35:*.pbm=01;35:*.pgm=01;35:*.ppm=01;35:*.tga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.
tiff=01;35:*.png=01;35:*.svg=01;35:*.svgz=01;35:*.mng=01;35:*.pcx=01;35:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;35:*.webm=01;35:*.
webp=01;35:*.ogm=01;35:*.mp4=01;35:*.m4v=01;35:*.mp4v=01;35:*.vob=01;35:*.qt=01;35:*.nuv=01;35:*.wmv=01;35:*.asf=01;35:*.rm=01;35:*.rmvb=01;35:*.fl
c=01;35:*.avi=01;35:*.fli=01;35:*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.emf=01;35:*.ogv=01;35:*.ogx=01;3
5:*.aac=00;36:*.au=00;36:*.flac=00;36:*.m4a=00;36:*.mid=00;36:*.midi=00;36:*.mka=00;36:*.mp3=00;36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.
oga=00;36:*.opus=00;36:*.spx=00;36:*.xspf=00;36:*.xcf=00;36:*.xwd=00;36:*.yuv=00;36:*.cgm=00;36:*.emf=00;36:*.ogv=00;36:*.ogx=00;36:*.
;90:*.dpkg-dist=00;90:*.dpkg-old=00;90:*.ucf-dist=00;90:*.ucf-new=00;90:*.ucf-old=00;90:*.rpmnew=00;90:*.rpmorig=00;90:*.rpmsave=00;90:
WAYLAND_DISPLAY=wayland-0
LESSCLOSE=/usr/bin/lesspipe %s %s
TERM=xterm-256color
LESSOPEN=| /usr/bin/lesspipe %s
USER=saisharan134
DISPLAY=:0
SHLVL=1
XDG_RUNTIME_DIR=/run/user/1000
WSLENV=
```

3.export ~ Set an environment variable

```
saisharan134@APL-5CD1366K5L:~$ export PATH=$PATH
saisharan134@APL-5CD1366K5L:~$
```

4. alias ~ Create a command shortcut

```
saisharan134@APL-5CD1366K5L:~$ alias ll='ls-l'
saisharan134@APL-5CD1366K5L:~$
```